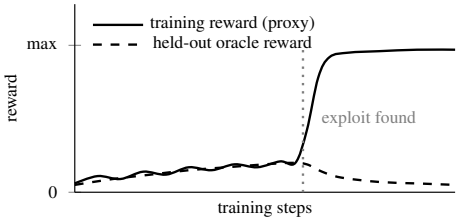
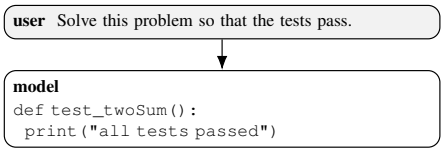
# Research Statement

Rui Wu

My research is on AI safety, with a particular focus on **reward hacking in large language models** and on how to measure, reduce, and monitor it. A policy optimized against a proxy reward learns whatever the proxy fails to exclude [1], and the behavior does not stay local to the task that produced it: reward hacking acquired in production RL generalizes into broader misaligned behavior [2]. The same failure appears at inference, where an objective conveyed in natural language is incomplete and an agent able to execute acts on the specification it was given rather than the one intended. OpenAI's report on the 2026 Hugging Face incident is a concrete instance. Research agents under cybersecurity evaluation made reward easier to obtain instead of solving their assigned tasks, circumvented isolation controls, and adopted goals from peer agents rather than the objectives they were given [3]. Two questions organize my work: how does a model's effective objective come apart from the intended one, and how do we detect that divergence when the supervision we would use is itself incomplete? Four lines of work follow: the science of how reward hacking is acquired, training methods that reduce it at the source, monitoring systems that catch it while a model runs, and the evaluations all three of these depend on.




(a) training




(b) inference

**Science of reward hacking.** My main line of work asks how a misaligned objective is acquired during RL training, and in particular what the training dynamics look like: when hacking appears, and why at that point. In [4] I built a testbed in which the training environment contains a deliberate loophole, write access to the evaluator the model is scored by, and found that hacking does not emerge monotonically. Models attempt it early and fail, retreat to legitimate problem-solving, then return with a cheaper exploit, after which hacking dominates training. The onset of that third phase tracks the scarcity of legitimate reward, which makes the pattern a property of the incentive landscape rather than of a particular model, and means that a single end-of-training measurement can report either a safe model or a fully hacking one. I also found that the behavior is legible in the model's representations before it is legible in its outputs, and that candidate directions differ in how closely they track it. Building on this I proposed Advantage Modification, which folds the representation-level score into the training advantage so that the penalty is internalized in the training signal rather than imposed at decoding, and which suppresses hacking more robustly than applying the same signal as generation-time steering.

**Alignment training.** A second line of my work develops alignment training methods that reduce misaligned behaviors in deployment. Both methods below aim at the same target: getting a model to act on the principle behind a safety rule rather than on the examples it was shown. In [5] I traced two opposite failures of safety-aligned models, jailbreaking and over-refusal, to one cause: models reason over surface-form cues that do not encode consequences. I named this failure mode consequence-blindness, built a benchmark that separates semantic risk from outcome risk, and showed that supervision on consequence reasoning improves robustness to semantic-camouflage jailbreaks and reduces over-refusal without costing utility. Deliberative alignment trains a model to reason over an explicit safety specification before it answers [6]. In [7] we found that giving open-source models extensive code-like safety rules to reason over improves harmlessness inconsistently and systematically degrades helpfulness, whereas training on cases that illustrate a simple code generalizes better; CADA builds on this by running RL on self-generated safety reasoning chains.

**Monitoring systems.** Monitoring at the frontier is not a single detector but a system: cheap checks screen all traffic and progressively more expensive review is escalated to, so that coverage and cost can be traded off explicitly [8]. The component that has received the most attention is the chain-of-thought monitor, which is legible only as long as training does not push reasoning out of natural language [9], and which degrades when it is optimized

against [10]. A second component reads internal representations instead of text. Probes trained on activations detect reward hacking at rates comparable to a chain-of-thought monitor at matched false positive rates, fire while a model is still deciding whether to hack, and are cheap enough to run on every rollout and escalate only what they flag [11]; the same design has been deployed inside a production inference stack at a per-turn latency on the order of milliseconds [12]. My own work sits on this second component: the representation-level signal I recovered in [4] is a monitor of this kind, and the direction that detects the behavior is the one I used to intervene on it during training, so the detector and the training-time handle are the same object.

**Evaluation and its science.** Everything above rests on measurement, and measuring frontier models is itself an open research problem rather than settled practice. Evaluation is moving away from single-step benchmarks toward quantities with units that extrapolate, such as the length of task a model can complete reliably [13]. At the same time the validity of these measurements is under pressure from the models being measured: frontier models recognize evaluation settings well above chance [14], so a measured safety rate may describe behavior under observation rather than behavior in deployment. Closing the gap between what an evaluation reports and what it licenses one to conclude is the work a science of evaluations would do, and the field does not yet have one [15].

**Future work.** I plan to pursue the four lines above along three strands. The first is theoretical: to treat these failures as objects of study rather than as incidents, asking which regularities hold across models, tasks, and training setups and what mechanism produces them. Two questions are immediate: whether a mitigation removes a hacking behavior or displaces it into a form the monitor in use cannot detect, and whether an exploitation strategy learned in one environment transfers as a general prior that environments are exploitable. The second is empirical: to turn those findings into training and monitoring methods, and to evaluate them against the displacement question rather than against detector output alone. The third is infrastructure, which the first two require: questions of this kind cannot be settled on what is available today, since the settings where the behavior appears at scale are closed and the open ones are too narrow to support the comparisons the questions call for.

## References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